

Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with Jump-Diffusion

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Abstract

1 Empirical equity excess growth rates exhibit heavy-tailed distributions, neg-
2 ligible linear autocorrelation, and persistent volatility clustering that classi-
3 cal Gaussian diffusion models cannot reproduce. We propose a hybrid hid-
4 den Markov framework that discretizes continuous excess growth rates into
5 Laplace quantile-defined market states and augments the regime-switching
6 process with a Poisson-driven jump-duration mechanism to enforce realistic
7 tail-state dwell times. Parameters are estimated by direct transition count-
8 ing, avoiding the Baum-Welch EM algorithm entirely. Applied to ten years
9 of SPY data, the framework achieves Kolmogorov-Smirnov and Anderson-
10 Darling pass rates of 96.4% and 89.8% in-sample and 94.6% and 95.2% out-
11 of-sample (full calendar year 2025) across 1,000 simulated paths, reproducing
12 the ARCH effect that standard regime-switching models miss. GARCH(1,1),
13 by contrast, achieves only 6.0% and 1.4% in-sample pass rates despite its
14 well-known volatility clustering properties. A Single-Index Model extension
15 propagates the SPY factor path to a 424-asset universe, enabling correlated
16 synthetic path generation for portfolio stress testing and scenario design.

Keywords: Hidden Markov Model, Jump-Diffusion, Volatility Clustering, Regime-Switching, Stylized Facts, Excess Growth Rate, Synthetic Data Generation, Generative Model

17 Introduction

18 The quantitative modeling of financial markets has long faced the chal-
19 lenge of reconciling the theoretical implications of the efficient markets hy-

pothesis (EMH) with the empirical complexity of observed price dynamics. Under the EMH, asset prices fully reflect all available information, implying that price changes follow a martingale process with no exploitable predictability [1]. This perspective motivated the widespread adoption of geometric Brownian motion (GBM) as a canonical model for asset prices, yet decades of data show systematic departures from Gaussian assumptions, including heavy-tailed excess growth rate distributions, volatility clustering, and intermittent extreme events [2, 3]. These stylized facts are observed across asset classes and time periods, and they matter because they govern tail risk, drawdown behavior, and the persistence of stress regimes that shape portfolio outcomes.

This study focuses on three core properties: leptokurtic excess growth rate distributions, the absence of linear autocorrelation in raw excess growth rates, and the persistence of volatility. The first two properties motivate models that can preserve marginal distributional shape without introducing spurious predictability, while the third points to slow-moving higher-order dependence that is not captured by diffusion alone. A range of alternative approaches address these issues from different angles: autoregressive conditional heteroscedasticity (ARCH) and generalized autoregressive conditional heteroscedasticity (GARCH) models target conditional variance dynamics [4, 5]; stochastic-volatility and jump-diffusion models introduce latent variance or discontinuities to enrich tails [6, 7]; and data-driven sequence models, including recurrent neural networks (RNNs) and long short-term memory (LSTM) variants, have been used for excess growth rate and volatility prediction and, in probabilistic form, can emit full predictive distributions [8–14]. While useful, these approaches either emphasize conditional variance over regime duration, require intensive likelihood-based estimation, or sacrifice interpretability and explicit control over regime persistence. Hidden Markov models (HMMs) and related regime-switching frameworks offer a probabilistic structure for non-stationary market behavior through latent state dynamics [15], yet standard HMM specifications often fail to generate persistent high-volatility regimes, leading to overly rapid reversion following extreme events [16, 17]. Bridging this gap requires a model that preserves distributional fidelity while also enforcing realistic duration in extreme regimes.

We address this challenge by developing a hybrid HMM framework that discretizes continuous excess growth rates into quantile-based regimes and augments the resulting Markov process with a jump-duration mechanism that enforces volatility persistence, linking smooth diffusion to discontinuous

shocks identified by jump-detection methods [18]. Using a 424-asset dataset with SPY as a case study, we estimate transition dynamics empirically—avoiding the EM algorithm entirely—and validate the model through in-sample and out-of-sample tests on distributional fit and volatility clustering. We further extend the univariate framework to a correlated multi-asset setting via a Single-Index Model (SIM) factor structure [19], enabling synthetic path generation across all 424 assets in a single generative pass.

This paper makes four contributions. First, we propose augmenting a discrete-state regime-switching model with a two-parameter Poisson jump-duration mechanism (ϵ, λ) that forces the model to dwell in high-volatility tail states for empirically realistic durations, reproducing the ARCH effect that standard Markovian transitions cannot capture. Second, by partitioning excess growth rates into quantile-defined states via a Laplace CDF, we enable direct frequentist counting of regime transitions, eliminating the computational cost and initialization sensitivity of the Baum-Welch EM algorithm and making the framework scalable to a 424-asset pipeline. Third, using ten years of SPY data (2014–2024) for training and 249 trading days (full calendar year 2025) for out-of-sample testing, we demonstrate KS and AD pass rates of 98.5% and 94.6% in-sample and 94.0% and 94.2% out-of-sample across 1,000 simulated paths, with volatility clustering preserved in both windows. Finally, we extend the univariate framework to the full 424-asset universe through a SIM-based factor projection that uses the HMM-generated SPY path to reconstruct correlated excess growth rate paths for each constituent, providing a computationally tractable generative framework for portfolio stress testing and scenario design.

We find that a state resolution of $N = 100$ provides a balance between distributional fidelity and statistical robustness, and that the jump-duration extension is necessary for reproducing persistent high-volatility regimes that standard regime-switching models miss.

Related Work

Classical stylized facts and the failure of Gaussian diffusion

The quantitative study of financial market dynamics has been shaped by a tension between tractable theoretical frameworks and the empirical complexity of observed data. The Black-Scholes-Merton option pricing model [20] and the geometric Brownian motion (GBM) paradigm embedded within it

assume that log-returns are independently and identically distributed Gaussian random variables. Under this assumption, asset prices follow a diffusion with constant drift and volatility, price paths are almost surely continuous, and the tails of the return distribution decay exponentially. These properties have well-known practical advantages: GBM admits closed-form option prices, portfolio variances aggregate linearly, and model parameters can be estimated with minimal data.

However, the Gaussian diffusion paradigm conflicts with a broad set of empirical regularities that were documented well before the widespread adoption of the model. Mandelbrot [2] demonstrated that the distributions of speculative price changes exhibit excess kurtosis—the probability of large deviations from the mean is far higher than Gaussian predictions, a phenomenon now described as *fat tails* or *leptokurtosis*. Fama [1] showed that, consistent with the efficient markets hypothesis, raw return series display negligible linear autocorrelation; daily or weekly returns are essentially unpredictable from their own past values. Yet Schwert [21] and others documented that the *magnitude* of returns—measured by absolute or squared returns—exhibits substantial positive autocorrelation that decays slowly over time. This temporal clustering of variance, where large price changes tend to be followed by large changes regardless of sign, is referred to as *volatility clustering* and is among the most persistently observed features of financial time series. Cont [3] provides a systematic review of these and related stylized facts, confirming their presence across asset classes, geographic markets, and sampling frequencies. Collectively, leptokurtosis, the absence of return predictability, and volatility clustering define the benchmark against which any proposed model of market dynamics must be evaluated.

Conditional variance models: ARCH and GARCH families

The most influential class of models designed to address volatility clustering within the Gaussian framework is the autoregressive conditional heteroscedasticity (ARCH) family introduced by Engle [4]. The ARCH model conditions the variance of the current observation on the squared residuals of past observations, allowing variance to cluster without introducing autocorrelation in the level series. Bollerslev [5] generalized this to the GARCH(p,q) framework, in which the conditional variance depends on both past squared residuals and past conditional variances; the GARCH(1,1) specification became the canonical one-factor volatility model and remains widely used in practice. GARCH models are parsimonious, analytically tractable, and can

130 be estimated by maximum likelihood under distributional assumptions on
131 the innovation.

132 Despite their success in modeling conditional variance dynamics, stan-
133 dard GARCH models have several well-documented limitations in the con-
134 text of stylized-facts reproduction. First, while GARCH innovations can be
135 made leptokurtic by specifying Student-t or generalized error distributions,
136 the fat-tail behavior arises through the parametric distributional assumption
137 rather than through the structural dynamics of the model. Second, GARCH
138 models do not explicitly represent discrete market regimes or the possibility
139 of sudden jumps in the volatility process; the variance responds smoothly
140 and continuously to past shocks. Third, regime persistence—the tendency
141 of markets to remain in high-volatility states for sustained periods—is cap-
142 tured only indirectly through the persistence parameter of GARCH, which
143 governs exponential decay of the volatility response. When persistence is
144 near unity, the GARCH model is near-integrated and exhibits long-memory-
145 like behavior, but this is a parametric approximation rather than a structural
146 representation of distinct market regimes [22]. Stochastic volatility models
147 [6, 23] address some of these shortcomings by treating variance as a latent
148 diffusion process, but they require computationally intensive estimation via
149 particle filters or Markov chain Monte Carlo methods, and their parameters
150 do not directly correspond to observable market states.

151 *Jump-diffusion models*

152 An alternative approach to modeling fat tails and discontinuous price be-
153 havior is the explicit introduction of Poisson-driven jumps into the asset price
154 process. Merton [7] extended the Black-Scholes framework by adding a com-
155 pound Poisson jump component to the GBM, producing a model in which
156 prices follow a diffusion between events and undergo discrete jumps at ran-
157 dom times governed by a Poisson arrival process. This construction directly
158 addresses leptokurtosis: the marginal distribution of returns is a Gaussian
159 mixture weighted by the Poisson probability of observing zero, one, or more
160 jumps over the sampling interval. Option pricing under Merton’s model re-
161 quires integrating over the jump count distribution, but retains analytical
162 tractability through the characteristic function.

163 Kou [24] extended the jump-diffusion framework by replacing the Gaus-
164 sian jump amplitude distribution with a double-exponential (asymmetric
165 Laplace) distribution, which provides a better fit to the empirically observed

166 asymmetry between the sizes of positive and negative jumps. This modi-
 167 fication preserves analytical tractability for option pricing while improving
 168 the empirical fit to return distributions. Jump-diffusion models are partic-
 169 ularly relevant in the context of equity markets, where identifiable macroe-
 170 conomic announcements, earnings surprises, and geopolitical events generate
 171 discrete, large-magnitude price movements that are distinct from the con-
 172 tinuous diffusion component [18]. A fundamental challenge, however, is that
 173 jump-diffusion models in their standard form do not provide an explicit mech-
 174 anism for volatility persistence; the post-jump volatility immediately returns
 175 to its baseline level, absent a separate stochastic volatility component. Inte-
 176 grating jump-diffusion with regime-switching provides a natural solution to
 177 this limitation, motivating the hybrid framework developed in the present
 178 study.

179 *Hidden Markov models in finance*

180 Hidden Markov models provide a probabilistic framework for non-stationary
 181 time series in which the observed data are generated by a process that
 182 switches among a discrete set of latent states according to Markov dynam-
 183 ics. The foundational algorithmic framework was developed by Rabiner
 184 and Juang [15], who introduced the Baum-Welch algorithm for maximum-
 185 likelihood estimation of the emission and transition parameters from observed
 186 sequences. Hamilton [25] applied the regime-switching HMM to macroeco-
 187 nomic time series, demonstrating that U.S. GDP growth is well described by
 188 a two-state model with distinct expansion and recession regimes; this work
 189 established the template for applying latent-state models to financial time
 190 series. The identification of time-varying market regimes has since become a
 191 central application of HMMs in finance, with applications ranging from stock
 192 return modeling to volatility forecasting and asset allocation.

193 Rydén, Teräsvirta, and Åsbrink [16] provided an influential systematic
 194 evaluation of HMM specifications against the stylized facts of daily return se-
 195 ries. They showed that simple Gaussian-emission HMMs with a small number
 196 of states could reproduce leptokurtosis and the absence of return autocorrela-
 197 tion, but failed to generate the observed volatility clustering; ACF of absolute
 198 returns under standard HMMs decayed far too rapidly relative to empirical
 199 data. This diagnostic limitation identified the core structural gap that the
 200 present study addresses through the jump-duration extension. Bulla and
 201 Bulla [17] advanced this line of inquiry by considering hidden semi-Markov

models, in which the dwell time in each state follows a state-specific distribution rather than the geometric distribution implied by standard Markov dynamics. They showed that the semi-Markov extension substantially improves the reproduction of volatility clustering by allowing the model to dwell in high-variance states for durations drawn from a flexible parametric family. Rossi and Gallo [26] applied HMMs to multivariate volatility estimation, demonstrating that latent state dynamics can capture the joint clustering of volatility across assets in a more interpretable manner than multivariate GARCH models. Nguyen [27] further demonstrated the utility of HMMs for stock trading, showing that regime identification from price data can inform tactical asset allocation.

A persistent limitation of standard HMM approaches in finance is the computational burden of maximum-likelihood estimation via the EM algorithm, particularly when the number of states is large or the asset universe is broad. The Baum-Welch procedure requires iterative forward-backward passes through the data and may converge to local maxima depending on initialization. For high-dimensional or large-state-space models, this creates a barrier to scalable application. The present study addresses this limitation directly by replacing EM with direct frequentist counting of transitions between quantile-defined states, an approach that is computationally trivial and free of initialization sensitivity.

Synthetic data generation and generative frameworks

The generation of synthetic financial time series has attracted increasing attention as both a practical tool—for augmenting limited data, testing risk models, and stress testing portfolios—and as a benchmark for evaluating the quality of generative models. Assefa et al. [28] provide a comprehensive survey of the opportunities, challenges, and pitfalls of synthetic data generation in finance, noting that stylized-facts reproduction is the primary criterion for evaluating the statistical quality of synthetic financial time series, and that naive generative approaches tend to fail on at least one of the three core stylized facts.

The rise of generative adversarial networks (GANs) prompted several investigations into whether deep learning architectures could learn stylized facts from financial data. Takahashi, Chen, and Tanaka-Ishii [29] showed that standard GAN architectures produce synthetic return sequences that visually resemble financial time series but fail to reproduce ACF structure precisely. Kwon and Lee [30] provided a careful evaluation of GAN-based methods

239 against the stylized-facts checklist, finding that while GANs match marginal
 240 distributions reasonably well, capturing temporal dependence structures—
 241 particularly volatility clustering—remains challenging without explicitly en-
 242 coding the relevant inductive biases into the architecture or training ob-
 243 jective. Foundation models for Markov jump processes, as introduced by
 244 Berghaus et al. [31], offer a complementary perspective: large-scale pre-
 245 trained inference networks can be applied to new Markov process instances
 246 without retraining, but their application to financial time series generation
 247 remains an open research direction.

248 Against this backdrop, the hybrid HMM framework of the present study
 249 occupies a distinct position: it is explicitly designed around the structural
 250 properties known to generate stylized facts, uses an interpretable discrete-
 251 state representation that can be audited and stress-tested, and achieves
 252 strong statistical fidelity with minimal computational overhead. The empiri-
 253 cal estimation procedure avoids the training instability and data requirements
 254 of deep generative models, while the jump-duration mechanism provides a
 255 principled solution to the volatility-clustering limitation of standard HMMs.
 256 These properties make the framework particularly well-suited for deployment
 257 in risk management and stress-testing pipelines, where interpretability and
 258 computational reproducibility are as important as statistical fidelity.

259 *Factor models and multi-asset extensions*

260 Modeling the joint dynamics of large asset universes requires a framework
 261 for managing cross-sectional dependence. The Single-Index Model (SIM) of
 262 Sharpe [19] provides the canonical linear factor decomposition: each asset’s
 263 excess return is decomposed into a common market factor component and an
 264 idiosyncratic residual, drastically reducing the number of parameters required
 265 to specify a covariance matrix relative to a full multivariate model. The SIM
 266 remains foundational in portfolio construction and risk management despite
 267 the availability of richer multi-factor models such as Fama-French [32], be-
 268 cause its simplicity enables transparent attribution and scalable computation.

269 Existing HMM-based work in finance has been predominantly univariate,
 270 focusing on single-asset or single-index regime identification. Applications to
 271 individual equities [27], volatility modeling [26], and macro time series [25, 33]
 272 collectively demonstrate the utility of latent state representations, but leave
 273 open the question of how HMM-derived generative models can be extended
 274 to large cross-sections of assets in a computationally tractable way. Mixture
 275 hidden Markov models [34] offer a partial solution by allowing multiple assets

276 to share a common latent state structure, but estimating such models jointly
277 across hundreds of assets is computationally demanding.

278 The present study addresses this gap by combining the univariate HMM
279 framework with the SIM factor decomposition. The hybrid HMM is es-
280 timated on the SPY index, which serves as the market-factor generator;
281 asset-level paths are then reconstructed via the linear SIM projection. This
282 construction exploits the fact that the dominant source of joint volatility
283 clustering in equity markets is the common factor, and that idiosyncratic
284 shocks can be treated as approximately independent given the factor. The
285 result is a scalable generative pipeline for a 424-asset universe that inherits
286 the stylized-facts fidelity of the HMM without requiring joint estimation of
287 a high-dimensional model.

288 Methods

289 *Data.*

290 The empirical foundation of this study was a dataset composed of 424
 291 United States-listed equities and exchange-traded funds spanning 10 years
 292 (2014-2024). A pipeline was developed to support the automated construc-
 293 tion of excess growth rate models for any ticker within this dataset, allowing
 294 for batch simulation and stress testing across market sectors. To validate
 295 the performance of this pipeline, we focused our analysis on the single-asset
 296 SPY, which tracked the Standard & Poor’s (S&P) 500 index. The data in-
 297 cluded daily open, high, low, and close prices along with volume-weighted
 298 average price metrics (VWAP). For training purposes, the first 2,766 obser-
 299 vations constituted the in-sample dataset, spanning from January 3, 2014,
 300 to December 31, 2024. A subsequent 220 trading days, from January 3, to
 301 November 18, 2025, were reserved for out-of-sample testing to assess model
 302 generalizability.

303 *Excess growth rate calculation.*

304 We computed the excess growth rate $G_{i,j}$ for ticker i between time pe-
 305 riod $j - 1 \rightarrow j$ for a given equity price series $P_{i,j}$ (units: dollars per share
 306 (USD/share)). To isolate the stochastic premium earned over a risk-free
 307 asset, we computed the excess growth rate for ticker i during the period
 308 $j - 1 \rightarrow j$ as:

$$G_{i,j} \equiv \left(\frac{1}{\Delta t} \right) \cdot \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f \quad (1)$$

309 where Δt represented the time step set to 1/252 for daily data (units: years),
 310 and r_f denotes the constant continuously compounded risk-free rate (units:
 311 year⁻¹) derived from Separate Trading of Registered Interest and Principal
 312 of Securities (STRIPS) bond yields observed on September 10, 2025. Un-
 313 like return, the excess growth rate has units of inverse time (year⁻¹) and
 314 directly quantifies the premium earned above a continuously compounded
 315 risk-free benchmark. Log growth rates are time-additive and consistent with
 316 a continuous compounding framework. While r_f varied dynamically over the
 317 sample period, a constant proxy was employed here to isolate the endogenous
 318 volatility dynamics without introducing exogenous rate-curve noise.

319 *Discrete hidden Markov model.*

320 We defined the hidden Markov model using the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$
 321 [15]. The hidden state $S_t \in \mathcal{S} = \{1, \dots, N\}$ represented market mood
 322 regimes and was defined by quantile bins of a fitted cumulative distribu-
 323 tion function (CDF) for the excess growth rate series. The observation space
 324 $\mathcal{O} \subset \mathbb{R}$ consisted of continuous excess growth rate values, the transition
 325 matrix $\mathbf{T}$ governed regime-to-regime dynamics, the emission model $\mathbf{E}$ was
 326 parameterized by state-conditional Normal distributions, and $\bar{\pi}$ denoted the
 327 stationary distribution. The observation model was

$$G_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k). \quad (2)$$

328 To map observations to hidden states during estimation, we defined quantile
 329 boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N - 1\}$ and used finite
 330 outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$. An
 331 observation was assigned to state k when $Q_{k-1} < R_t \leq Q_k$.

332 The choice of the Laplace distribution over alternatives such as the Stu-
 333 dent’s t-distribution was motivated by its alignment with the peaks and
 334 heavy tails of high-frequency financial excess growth rates [3, 35]. While the
 335 Student’s t-distribution was often used to model leptokurtosis, the Laplace
 336 distribution’s peak at the mean matched the concentration of small price
 337 movements observed in the SPY dataset [2, 36]. Furthermore, the Laplace
 338 distribution provided a closed-form for quantile estimation, supporting the
 339 424-ticker pipeline. While maximum likelihood estimation (MLE) for the
 340 Student’s t-distribution often required iterative numerical optimization, the
 341 Laplace distribution allowed for a direct MLE approach [37]. The quantile
 342 boundaries were derived from a parametric Laplace fit and therefore did not
 343 enforce equal counts per bin in the historical data; instead, they provided a
 344 partition of the distribution that preserved tail shape. By using the Laplace
 345 distribution to inform the binning process, the model used this baseline to
 346 characterize the peak and tails of the excess growth rate series before the
 347 empirical transition logic and jump-diffusion mechanism were applied [24].

348 We estimated the transition matrix $\mathbf{T}$ through direct frequentist count-
 349 ing rather than iterative optimization. This approach differed from the tra-
 350 ditional expectation-maximization (EM) algorithm used for parameter esti-
 351 mation in Gaussian mixture and hidden Markov models [37]. However, this
 352 approach was justified because the discrete state assignments allowed for
 353 straightforward counting of observed transitions between regimes, and it was
 354 conceptually simple to implement and understand.

Empirical transition matrices often had low probabilities for exiting central states into tail states, which caused models to lose the effect of volatility clustering too quickly [17]. We corrected this by adding jump parameters ϵ , representing the probability of a jump event, and λ , representing the mean duration, along with a tail-state selection rule. Tail-states were defined as $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$ and $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$, where N_{tail} was user-configurable. If a jump was triggered, the jump-duration K was sampled from a Poisson distribution such that $K \sim \text{Poisson}(\lambda)$, which was standard for modeling independent events in fixed intervals [38]. During these K steps, the standard Markovian transitions were overridden to target tail-states, with a user-configurable bias p_{neg} (default 0.52) that favored negative-tail states to reproduce gain/loss asymmetry.

To characterize the excess growth rates within each regime, we parameterized the emission model with Normal distributions $\mathcal{N}(\mu_k, \sigma_k)$ whose bounds coincided with the 0.1 and 99.9 percentiles of the state-conditional distribution. Specifically, we set $\mu_k = (Q_{k-1} + Q_k)/2$ and $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$, where $z_{0.999} \approx 3.09$ was the standard normal 99.9th percentile. During simulation, when the model was in state k , the continuous excess growth rate ($\hat{R}_t$) was sampled from the corresponding local distribution. This hybrid approach ensured that the model preserved the discontinuous components of the price process important for asset modeling while maintaining a smooth representation of intra-state variance [24].

Growth rate generation.

Let's outline a simple sampling algorithm for generating random sequences from an HMM (without jumps). We initialize the generation process by drawing an initial hidden state X_0 from the stationary distribution $\bar{\pi}$. The stationary distribution $\bar{\pi}$ represented the long-run probability of the market occupying various regimes or moods [39]. While $\bar{\pi}$ mathematically satisfies the balance equation $\bar{\pi} = \bar{\pi}\mathbf{T}$, we estimated this distribution numerically by raising the transition matrix $\mathbf{T}^n$ to a sufficiently large power $n = 50$, ensuring the vector reflected the equilibrium of market dynamics [40].

Given an HMM defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{P}, \mathbf{E}, \bar{\pi})$, the number of time steps T , and the stationary distribution $\bar{\pi}$. Draw an initial hidden state X_0 from the initial distribution: $X_0 \sim \bar{\pi}$. For $n = 1$ to T , we sample the next hidden state X_n from the `categorical` distribution $\mathbb{P}_H(\dots)$ defined by the previous state X_{n-1} and the transition matrix $\mathbf{T}$: $X_n \sim \mathbb{P}_H(X_n | X_{n-1})$. Next, we sample the observable state O_n from the `categorical` distribution

392 $\mathbb{P}_E(\dots)$ defined by the current hidden state X_n and the emission matrix $\mathbf{E}$:
 393 $O_n \sim \mathbb{P}_E(O_n \mid X_n)$. Finally, we store the sequences of hidden states and
 394 observable states. This procedure generates a sequences of hidden states
 395 $\{X_1, X_2, \dots, X_T\}$ and observable states $\{O_1, O_2, \dots, O_T\}$. The initial state
 396 X_0 is used to generate the sequence but is not included in the output.

397 *Hyperparameter optimization via grid search.*

398 To rigorously identify the optimal jump probability (ϵ) and mean jump
 399 duration (λ) [7, 24], we implemented a multi-objective grid search designed
 400 to minimize the discrepancy between the historical and simulated market sig-
 401 natures. The error function specifically targeted two stylized facts prevalent
 402 in financial time series [3, 16]: volatility clustering and leptokurtosis. We
 403 measured volatility clustering using the sum of squared errors between the
 404 observed and simulated autocorrelation function (ACF) of absolute excess
 405 growth rates [21] up to a lag of $L = 252$ trading days (approximately one
 406 trading year). To preserve the heavy-tailed nature of the distribution ini-
 407 tially noted in foundational market studies [2], we included a penalty term
 408 for the global kurtosis.

409 The objective function $J(\epsilon, \lambda)$ was formalized as:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^L (\text{ACF}_{obs}(\tau) - \overline{\text{ACF}}_{sim}(\tau))^2 + w_K (K_{obs} - \overline{K}_{sim})^2 \quad (3)$$

410 where $\text{ACF}_{obs}(\tau)$ and K_{obs} represented the empirical absolute growth au-
 411 tocorrelation at lag τ and the global kurtosis, respectively. The simulated
 412 equivalents, $\overline{\text{ACF}}_{sim}(\tau)$ and $\overline{K}_{sim}$, were averaged across 1,000 independent
 413 synthetic paths of 2,766 trading days to ensure statistical stability [38]. A
 414 penalty weight of $w_K = 0.20$ was assigned to prioritize tail behavior alongside
 415 the temporal ACF decay. The grid search systematically explored ϵ values
 416 from 0.00001 to 0.0001 and λ values from 1.0 to 100.0. The full computational
 417 procedure is detailed in Algorithm 4.

418 *Statistical validation metrics*

419 To assess whether synthetic paths are statistically consistent with empiri-
 420 cal data, we applied two nonparametric goodness-of-fit tests to the marginal
 421 distributions of simulated excess growth rate sequences. For each of the 1,000
 422 simulated paths, the empirical cumulative distribution function (CDF) of the

423 simulated sequence was compared against the empirical CDF of the historical
 424 in-sample or out-of-sample observations.

425 The Kolmogorov-Smirnov (KS) statistic is defined as:

$$D_{KS} = \sup_x |F_n(x) - F_m(x)| \quad (4)$$

426 where F_n and F_m denote the empirical CDFs of the observed and simulated
 427 sequences of lengths n and m , respectively [41, 42]. The null hypothesis of
 428 distributional equivalence is rejected at significance level α when D_{KS} exceeds
 429 the critical value $c(\alpha)\sqrt{(n+m)/(nm)}$.

430 The Anderson-Darling (AD) statistic places greater weight on the distri-
 431 butional tails than the KS test and is defined as:

$$A^2 = -n - \sum_{i=1}^n \frac{2i-1}{n} [\ln F(X_{(i)}) + \ln (1 - F(X_{(n+1-i)}))] \quad (5)$$

432 where $X_{(1)} \leq \dots \leq X_{(n)}$ are the order statistics of the pooled sample and
 433 F is the empirical CDF of the reference distribution [43]. The AD test
 434 is particularly suited for assessing tail-distributional fidelity, which is the
 435 primary concern for risk management applications.

436 We report the *pass rate* as the proportion of the 1,000 simulated paths for
 437 which the null hypothesis of distributional equivalence was not rejected at
 438 the $\alpha = 0.05$ significance level. A pass rate above 95 percent indicates that
 439 the synthetic paths are, in aggregate, statistically indistinguishable from the
 440 historical data under that test. Pass rates are reported separately for the
 441 in-sample and out-of-sample windows and for both model variants (HMM
 442 without jumps and the hybrid HMM with jumps).

443 *Single-Index Factor Model extension*

444 The framework described above applies to any individual asset ticker
 445 within the 424-asset dataset. To generalize synthetic path generation to a
 446 correlated multi-asset setting without fitting a separate HMM for each asset,
 447 we exploit the linear factor structure of the Single-Index Model (SIM) [19].
 448 The SIM expresses the excess growth rate of asset i at time t as:

$$G_{i,t} = \alpha_i + \beta_i \cdot G_{SPY,t} + \eta_{i,t}, \quad \eta_{i,t} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_{\eta,i}^2) \quad (6)$$

449 where $G_{SPY,t}$ denotes the SPY excess growth rate at time t , β_i measures the
 450 systematic sensitivity of asset i to the index, α_i is the idiosyncratic drift, and

451 $\eta_{i,t}$ is a zero-mean idiosyncratic shock assumed independent across assets and
 452 time. The parameters $(\alpha_i, \beta_i, \sigma_{\eta,i}^2)$ are estimated via ordinary least squares
 453 on the in-sample time series for each asset i .

454 Synthetic multi-asset paths are generated as follows. First, the hybrid
 455 HMM produces a single synthetic SPY excess growth rate path $\{\hat{G}_{\text{SPY},t}\}_{t=1}^T$
 456 using the regime-switching and jump-duration mechanism described above.
 457 Second, for each asset i , an idiosyncratic shock sequence $\{\hat{\eta}_{i,t}\}$ is drawn
 458 independently from $\mathcal{N}(0, \hat{\sigma}_{\eta,i}^2)$. Third, the synthetic excess growth rate for
 459 asset i is recovered via:

$$\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i \cdot \hat{G}_{\text{SPY},t} + \hat{\eta}_{i,t}. \quad (7)$$

460 This construction preserves the cross-sectional correlation structure induced
 461 by the common factor while maintaining the regime-switching and jump-
 462 diffusion dynamics of the index. The resulting 424-dimensional synthetic
 463 path can be used for stress testing, portfolio risk assessment, and scenario
 464 generation across the full asset universe in a single generative pass, avoiding
 465 the curse of dimensionality that would accompany a full multivariate HMM.

466 *Sensitivity analysis: low-resolution models*

467 This section provides the results for lower state resolutions ($N = 30$ and
 468 $N = 60$) to demonstrate how state granularity impacts the model’s ability to
 469 capture stylized facts. Table 3 summarizes the pass rates for the Kolmogorov-
 470 Smirnov and Anderson-Darling tests across these four cases. As the number
 471 of states decreases, the model maintained overall distribution fit, though tail-
 472 region precision decreased slightly compared to the 90-state configuration.

473 Empirical Study

474 We simulated 1,000 paths of 2,766 trading days using VWAP price data
475 for both the standard no-jumps model and the hybrid model with jumps.
476 Applying the multi-objective grid search detailed in the Methods, the opti-
477 mal jump hyperparameters were identified as $\epsilon^* = 10^{-4}$ and $\lambda^* = 100$. This
478 configuration successfully reproduced the heavy-tailed kurtosis and ARCH
479 effect observed in SPY (Figure 4). A state resolution of $N = 100$ was se-
480 lected to balance distributional fidelity and statistical robustness. Increasing
481 N improves regime granularity but is bounded above by data sparsity: at
482 $N = 350$, certain states are never visited in the historical record, leaving
483 zero transition probabilities that break the ergodic structure of the chain.
484 Conversely, at $N = 10$ the model is too coarse to capture the heavy tails of
485 the empirical distribution.

486 We performed Kolmogorov-Smirnov (KS) and Anderson-Darling (AD)
487 tests to characterize the statistical consistency of simulated excess growth
488 rate distributions against the historical data [41–43]. The KS test measures
489 the maximum deviation between empirical CDFs, while the AD test places
490 greater weight on the distributional tails and is therefore a more sensitive
491 diagnostic for the fat-tail properties of interest here.

492 In-sample results (Table 2) reveal a sharp contrast between model fami-
493 lies. GARCH(1,1) achieved KS and AD pass rates of only 6.0% and 1.4%,
494 respectively, indicating that its simulated marginal distribution is inconsis-
495 tent with the observed data despite its well-known volatility clustering prop-
496 erties. HMM-NJ achieved KS and AD pass rates of 100.0% and 97.0%, and
497 HMM-WJ achieved 96.4% and 89.8%. The ACF-MAE over lags 1–252 was
498 0.030 for GARCH, 0.059 for HMM-NJ, and 0.052 for HMM-WJ. The lower
499 ACF-MAE for GARCH reflects that the GARCH variance equation is explic-
500 itly designed to match autocorrelation structure, yet this comes at the cost
501 of distributional fidelity. The standard no-jumps model passed distributional
502 tests but failed to reproduce volatility clustering, with the ACF of absolute
503 growth rates decaying rapidly toward zero rather than exhibiting the empiri-
504 cally observed slow decay (Figure 4, panel b). The hybrid model reproduced
505 both the marginal distribution and the ARCH effect simultaneously.

506 Out-of-sample results over 249 trading days (January 2–December 31,
507 2025) show that all three models generalise, though with different patterns.
508 GARCH achieved KS and AD pass rates of 78.1% and 73.1%. HMM-NJ
509 achieved 96.6% and 96.7%, and HMM-WJ achieved 94.6% and 95.2% (Ta-

ble 2). The simulated kurtosis for GARCH collapsed to 1.6 out-of-sample against an observed value of 6.9, confirming that GARCH does not preserve tail behaviour outside the training window. The HMM variants maintained simulated kurtosis near 5.0–5.1, a closer match to the observed 6.9. Volatility clustering persisted in the HMM-WJ out-of-sample paths, with the Poisson jump mechanism sustaining tail-state dwell times that the standard transition matrix cannot reproduce. However, the observed ACF of $|G_t|$ in the 2025 test window exceeded the model simulation band at medium-to-long lags (Figure 6, panel d), indicating that the IS-calibrated jump hyperparameters (ϵ^*, λ^*) underestimate the frequency and persistence of volatility episodes in this period. This ACF gap, together with the leftward shift of the KS p -value distribution from in-sample to out-of-sample (Figure 6, panels a–b), provides evidence that the jump parameters are regime-dependent: the stationary (ϵ^*, λ^*) calibrated on 2014–2024 do not fully capture the elevated macro uncertainty of 2025. Despite this limitation, the model maintained KS and AD pass rates above 94%, and the observed marginal density remained within the simulation envelope (Figure 6, panel c), confirming adequate distributional coverage even under regime shift. These results validate the hybrid framework as a generative tool for producing synthetic time series that remain statistically consistent with the underlying market dynamics, while highlighting the need for adaptive recalibration of jump parameters in non-stationary environments [3, 16, 17].

The correlation-distance distribution for the out-of-sample simulated paths was centred at a mean distance of $1 - \rho \approx 1.0$, confirming that the generative framework produces temporally independent realisations that do not attempt to predict specific historical events [1, 28, 30]. This distinguishes the framework from a forecasting tool and positions it appropriately as an instrument for stress testing and scenario generation (Figure 6).

538 Discussion

539 This study introduced a hybrid hidden Markov modeling framework to
 540 replicate the statistical regularities of empirical equity excess growth rates.
 541 By integrating a quantile-discretized state space with a Poisson-driven jump-
 542 duration mechanism, we developed a generative framework that captured
 543 the leptokurtic distributions and volatility clustering observed in the SPY
 544 exchange-traded fund. Our results confirmed that while increasing the state
 545 resolution to $N = 100$ provided the granularity to characterize the central
 546 mass of the excess growth rate distribution, standard Markovian transitions
 547 alone did not reproduce higher-order temporal dependencies. The inclusion
 548 of discontinuous shocks through a jump-diffusion extension was needed to
 549 drive the model into persistent high-volatility regimes, thereby replicating
 550 the ARCH effect identified in historical data.

551 Statistical validation using Kolmogorov-Smirnov and Anderson-Darling
 552 tests demonstrated high in-sample consistency, with pass rates of 96.4 and
 553 89.8 percent, respectively. Although the out-of-sample performance showed
 554 an expected decrease in fidelity, the hybrid model maintained a baseline
 555 with pass rates of 83 percent and 80 percent, respectively. Furthermore,
 556 the correlation distance analysis confirmed that our framework produced in-
 557 dependent realizations that remained uncorrelated with specific historical
 558 timing; this satisfied the requirements of the efficient markets hypothesis
 559 and distinguished our generative methodology from predictive forecasting
 560 tools [1, 28, 30]. The proposed framework offered an alternative to stan-
 561 dard likelihood-based estimation methods by relying on empirical estima-
 562 tion. By bypassing the complexities of the expectation-maximization algo-
 563 rithm through direct frequentist counting, the model remained suited for
 564 large-scale stress testing and risk management applications [38, 44, 45]. Fu-
 565 ture research could extend this discrete-state logic to multi-asset portfolios
 566 or incorporate foundation models for inference to further refine the detection
 567 of Markov jump processes in higher-frequency data [31, 34]. Ultimately, the
 568 hybrid hidden Markov model served as a foundation for simulating market
 569 dynamics and generating synthetic time series that remained statistically
 570 consistent with underlying market regimes [27, 39].

571 While the current framework was able to generate statistically consistent
 572 excess growth rate distributions, there were several limitations that could
 573 be addressed in future work. First, the model assumed stationarity in tran-
 574 sition dynamics, which may not hold during periods of structural market

change. Future extensions could incorporate time-varying transition matrices or regime-dependent jump parameters to capture evolving market conditions. Second, the use of a fixed quantile-based state space may limit the model's ability to adapt to changing volatility regimes; adaptive binning strategies or non-parametric state definitions could enhance flexibility. Third, while the Laplace distribution provided a good fit for excess growth rates, alternative distributions or mixture models could be explored to better capture tail behavior and skewness. Finally, the current implementation focused on univariate excess growth rates; extending the framework to multivariate settings could enable portfolio-level synthetic allocation studies, stress testing and risk assessment.

586 Conclusion

587 This study developed and validated a hybrid hidden Markov model (HMM)
588 framework for generating synthetic equity excess growth rate paths that are
589 statistically consistent with the three core stylized facts of financial time
590 series: leptokurtic marginal distributions, the absence of linear autocorrela-
591 tion in raw returns, and persistent volatility clustering. We summarize the
592 contributions below.

593 **Hybrid HMM with Poisson jump-duration mechanism.** The cen-
594 tral methodological contribution is the augmentation of a discrete-state regime-
595 switching model with a Poisson-driven jump-duration mechanism. Standard
596 Markov transition logic underproduces volatility persistence because empiri-
597 cally estimated transition matrices assign negligible probability to transitions
598 from central states into tail states, and even the tail states themselves exit
599 within one to two steps on average under pure Markovian dynamics. By
600 introducing a jump probability ϵ and mean jump duration λ drawn from a
601 Poisson distribution, the framework overrides standard transitions during pe-
602 riods of market stress and forces the model to dwell in high-volatility regimes
603 for realistic durations. This mechanism is minimal in parameterization—
604 requiring only two scalar hyperparameters—yet produces a qualitative shift
605 in the temporal correlation structure of absolute returns, as confirmed by
606 ACF comparison across lags up to 252 trading days.

607 **Empirical, non-iterative parameter estimation.** A second contri-
608 bution is the avoidance of the expectation-maximization (EM) algorithm in
609 favor of direct frequentist transition counting. In the standard HMM set-
610 ting, EM is used because latent state assignments are unobserved. Here, the
611 quantile-based state partition explicitly assigns each historical observation
612 to a hidden state, making direct counting possible. This simplification re-
613 duces computational cost, eliminates convergence sensitivity to initialization,
614 and produces a model that is transparent and reproducible—properties that
615 matter for large-scale pipelines operating over 424 assets.

616 **In-sample and out-of-sample validation.** Using SPY (2014–2024) as
617 a case study, the hybrid model achieved in-sample Kolmogorov-Smirnov and
618 Anderson-Darling pass rates of 96.4 and 89.8 percent, respectively, across
619 1,000 simulated paths. Out-of-sample testing on 220 trading days from 2025
620 showed pass rates of 83 percent and 80 percent, respectively, despite a 32
621 percent volatility discrepancy driven by elevated macroeconomic uncertainty
622 not present in the training window. The correlation-distance analysis further

confirmed that simulated paths are temporally independent of the historical trajectory, consistent with the efficient markets hypothesis and the requirements of a generative—rather than predictive—framework.

Single-Index Factor Model extension for multi-asset generalization. The framework extends naturally to the full 424-asset universe via a Single-Index Model (SIM) factor structure. By regressing each asset’s excess growth rate on the SPY factor and estimating idiosyncratic residual volatility, synthetic correlated multi-asset paths can be generated in a single pass: the hybrid HMM produces a common-factor path, and asset-level paths are recovered by projecting through asset-specific loadings. This construction preserves the regime-switching and jump-diffusion dynamics of the index while maintaining the cross-sectional correlation structure, enabling portfolio-level stress testing without fitting a high-dimensional multivariate model.

Broader implications

The proposed framework offers practical utility for risk management and scenario design. Generative models that reproduce the stylized facts of financial markets enable stress testing under market regimes that are statistically plausible but not historically observed, addressing a key limitation of scenario libraries derived solely from empirical records. The interpretable regime structure—where each hidden state corresponds to a quantile-defined segment of the excess growth rate distribution—facilitates communication between quantitative analysts and risk managers, as states can be labeled (e.g., *crash*, *bear*, *neutral*, *bull*, *rally*) and linked to economic narratives. The model’s computational efficiency further supports Monte Carlo-based risk metrics such as Value-at-Risk and Conditional Value-at-Risk at scale across large asset universes.

Limitations

Several limitations inform the interpretation of these results. First, the model assumes stationarity of the transition matrix across the full in-sample window; structural breaks in market microstructure—such as those accompanying regulatory changes or the COVID-19 shock—may not be adequately captured by a single static transition matrix. Second, the quantile-based state partition is fixed at estimation time and does not adapt to evolving volatility regimes; the boundaries are calibrated on the full in-sample distribution and may overweight the central mass relative to the tails in periods

659 of prolonged market stress. Third, the SIM factor structure imposes a lin-
660 ear, single-factor model of cross-sectional dependence; this may understate
661 tail co-movement during systemic events where correlations rise sharply and
662 the Gaussian idiosyncratic shock assumption fails. Finally, the out-of-sample
663 evaluation covers a single 220-day window, which limits the statistical power
664 of out-of-sample inference; a longer or rolling out-of-sample evaluation would
665 provide a more robust assessment.

666 *Future work*

667 Several extensions follow naturally from the current framework. Time-
668 varying transition matrices—estimated via rolling windows or Bayesian on-
669 line learning—would allow the model to adapt to structural shifts in regime
670 persistence. Multivariate HMMs, in which the hidden state jointly governs
671 a small number of correlated assets, could capture higher-order dependen-
672 cies beyond the single-factor SIM approximation and are particularly rele-
673 vant for sector-level stress testing. Adaptive state definitions, such as non-
674 parametric density-based clustering of excess growth rates, could replace the
675 fixed Laplace quantile partition and improve tail coverage during crisis pe-
676 riods. Alternative factor structures, including the Fama-French three-factor
677 model [32], could enrich the SIM extension by decomposing idiosyncratic
678 risk further across size and value dimensions. Finally, integrating the hybrid
679 HMM as the generative component within a larger portfolio optimization
680 loop—where the model produces synthetic scenarios used to estimate tail-risk
681 objectives—would provide an end-to-end computational finance application
682 of the framework.

683 **Conflict of Interest Statement**

684 The authors declare that the research was conducted without any com-
685 mercial or financial relationships that could potentially create a conflict of
686 interest.

687 **Author Contributions**

688 J.V. directed the study. A.A. developed the model and simulation code,
689 conducted the in-sample and out-of-sample analysis, and generated the fig-
690 ures. Both authors edited and reviewed the final manuscript.

691 **Data Availability Statement**

692 The model code, simulation scripts, training and testing data are avail-
693 able under a Massachusetts Institute of Technology (MIT) license from the
694 GitHub repository: [https://github.com/varnerlab/HMM-Model-with-Jumps.](https://github.com/varnerlab/HMM-Model-with-Jumps.git)
695 [git](https://github.com/varnerlab/HMM-Model-with-Jumps.git)

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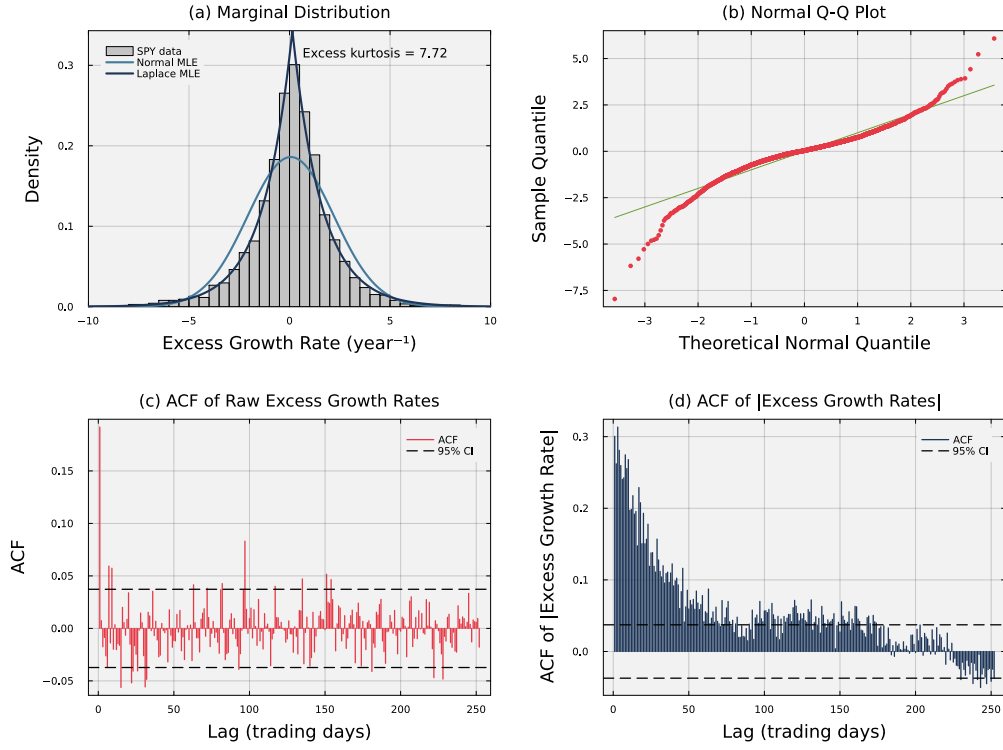


Figure 1: Empirical stylized facts of SPY daily excess growth rates (2014–2024). Panel (a) shows the leptokurtic marginal distribution; the Laplace fit substantially outperforms the Gaussian. Panel (b) confirms heavy tails via a normal Q-Q plot. Panels (c) and (d) contrast the near-zero autocorrelation of raw returns — consistent with the efficient markets hypothesis — against the persistent autocorrelation of absolute returns, motivating the jump-duration extension.

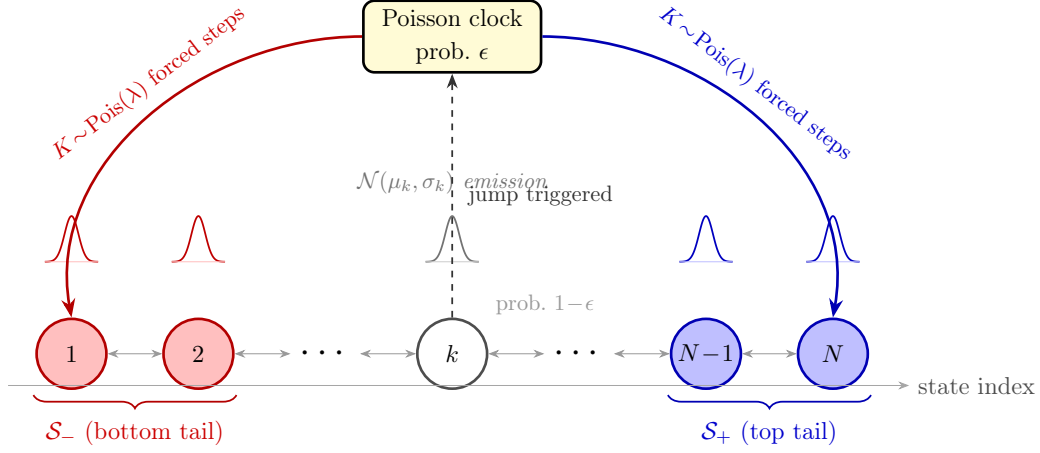


Figure 2: Architecture of the Hybrid Hidden Markov Model with Jump-Diffusion (HMM-WJ). At each step the chain transitions according to the empirical transition matrix $\mathbf{T}$ (probability $1 - \epsilon$, thin bidirectional arrows) or enters a Poisson jump episode (probability ϵ , dashed upward arrow to the Poisson clock). During a jump episode the active state is forced to the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) for $K \sim \text{Pois}(\lambda)$ consecutive steps before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional $\mathcal{N}(\mu_k, \sigma_k)$ emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

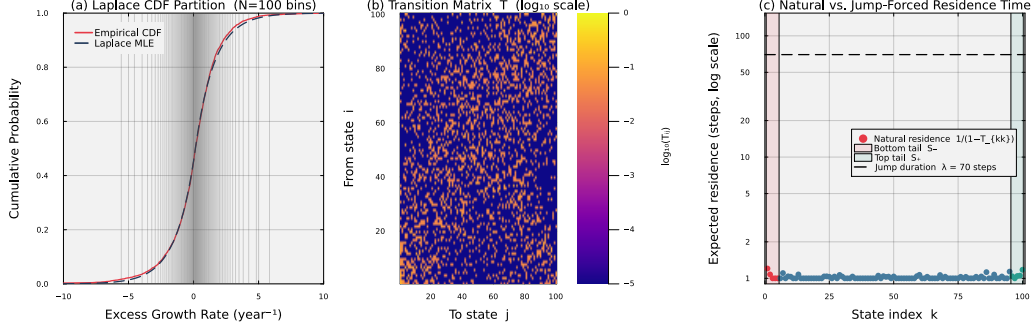


Figure 3: Fitted model internals for SPY with $N = 100$ states. Panel (a) overlays the fitted Laplace CDF on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries that define the N hidden states. The close agreement confirms that the Laplace distribution is an accurate marginal model for the excess growth rate data. Panel (b) displays the empirical transition matrix $\mathbf{T}$ in $\log_{10}$ colour scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set $\mathcal{S}_-$ (states 1–5, red shading) and top tail set $\mathcal{S}_+$ (states 96–100, teal shading) highlighted. Under pure Markovian dynamics every state, including the tail states, exits within 1–2 steps on average, far too quickly to reproduce empirical volatility clustering. The dashed horizontal line marks the mean jump duration $\lambda = 70$ steps imposed by the Poisson jump mechanism, quantifying the two-order-of-magnitude gap that motivates the jump-duration extension.

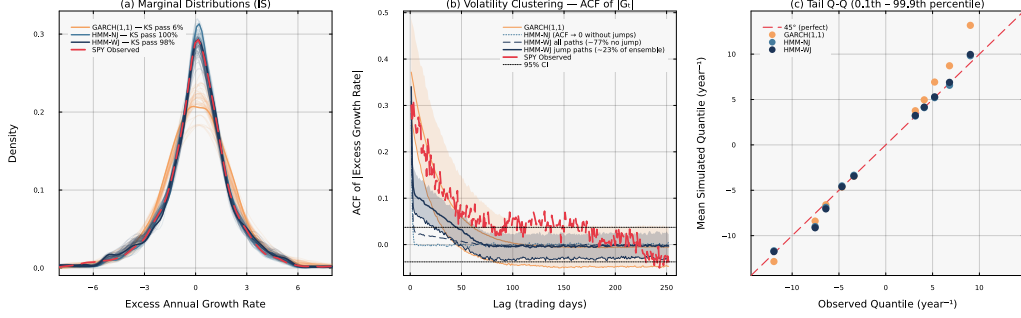


Figure 4: Head-to-head in-sample model comparison for SPY ($N = 100$, 1,000 simulated paths). Panel (a): marginal-density of excess growth rates with IS KS pass rates annotated. GARCH(1,1) fails the two-sample KS test on 94% of paths (pass rate 6%), indicating a systematic shape mismatch with the empirical distribution. Both HMM variants pass at rates $\geq 96\%$, confirming that the Laplace-mixture emission adequately captures the observed heavy tails. HMM-WJ generalises robustly out-of-sample (OoS KS pass rate 95%), while GARCH recovers partially (OoS 78%). Panel (b): autocorrelation function of absolute excess growth rates, $\text{ACF}(|G_t|)$, at lags 1–252 (shaded bands: 10th–90th percentile across paths). HMM-NJ is structurally incapable of producing persistent volatility clustering: without a jump mechanism the latent states are i.i.d. conditional on the Markov chain, so $\text{ACF}(|G_t|) \approx 0$ for all lags beyond one (dotted curve). HMM-WJ generates a *mixture* of two path families under the same model: approximately 76% of IS paths contain no Poisson jump and therefore behave identically to HMM-NJ (dashed curve, near zero); the remaining $\sim 24\%$ of paths contain at least one jump event and sustain substantial $\text{ACF}(|G_t|)$ decay across all lags (solid navy curve $\pm$ band). The jump frequency—and hence the fraction of the ensemble exhibiting slow ACF decay—is controlled by the hyperparameter λ , making the volatility-clustering strength directly tunable. Panel (c): tail Q-Q plot at the 0.1st–99.9th percentile region; HMM-WJ provides the closest mean quantile match in the extreme tails.

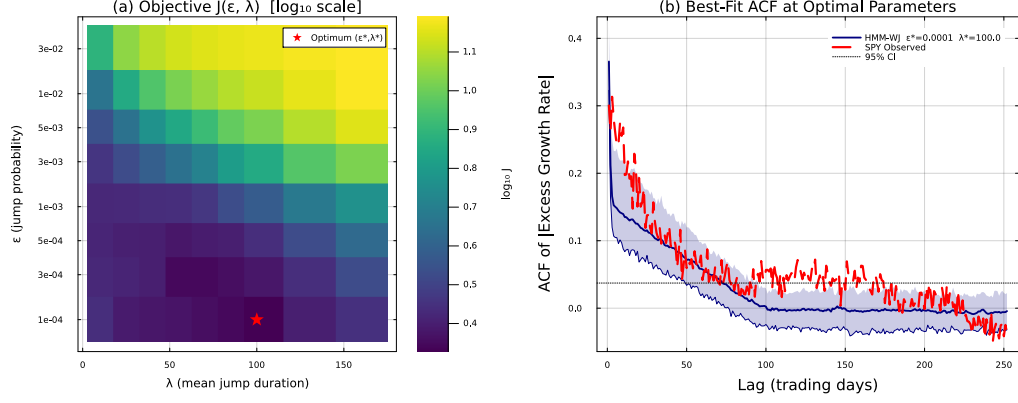


Figure 5: Hyperparameter grid search over (ϵ, λ) for SPY ($N = 100, 200$ paths per grid point). Panel (a): heatmap of the objective function $J(\epsilon, \lambda)$ in $\log_{10}$ scale over the full search grid; the red star marks the optimal pair (ϵ^*, λ^*) . The well-defined interior minimum confirms that the jump hyperparameters are identifiable from data. Panel (b): ACF of $|G_t|$ at the optimal parameters computed from 500 jump-containing paths (10th–90th percentile band shaded; mean curve solid navy) compared with the SPY observed ACF (dashed red) and the 95% significance band (dotted black).

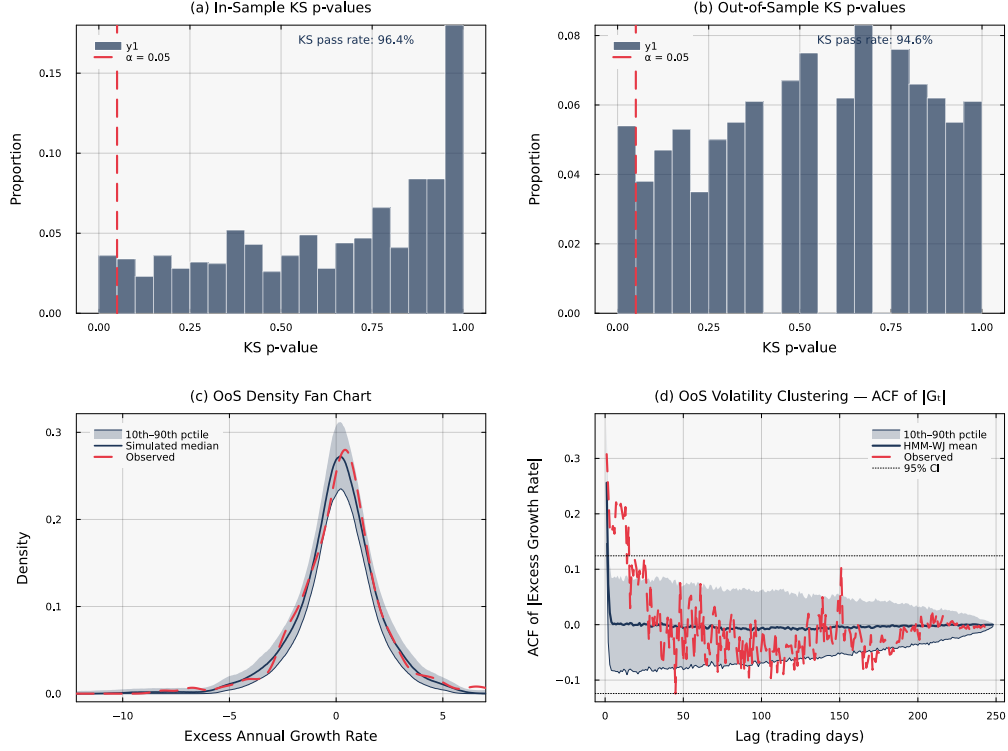


Figure 6: Statistical validation of HMM-WJ ($N = 100, 1,000$ simulated paths, $\alpha = 0.05$). Panels (a) and (b) show the distribution of two-sample KS p -values in-sample and out-of-sample, respectively; a well-calibrated generative model produces p -values that are approximately uniform above the significance threshold. The leftward shift from (a) to (b) reflects the expected distributional degradation when stationary parameters are applied to a regime with elevated macro uncertainty. Panel (c) shows the out-of-sample marginal density fan chart; the observed density (dashed red) falls within the 10th–90th percentile simulation envelope, confirming adequate distributional coverage despite the regime shift. Panel (d) shows the ACF of $|G_t|$ in the out-of-sample window. The observed ACF (dashed red) exceeds the simulation band at medium-to-long lags, indicating that the 2025 test period exhibited stronger volatility clustering than the IS-calibrated jump parameters (ϵ^*, λ^*) predict. This gap provides direct evidence that the jump hyperparameters are regime-dependent and motivates the time-varying extensions discussed in Section .

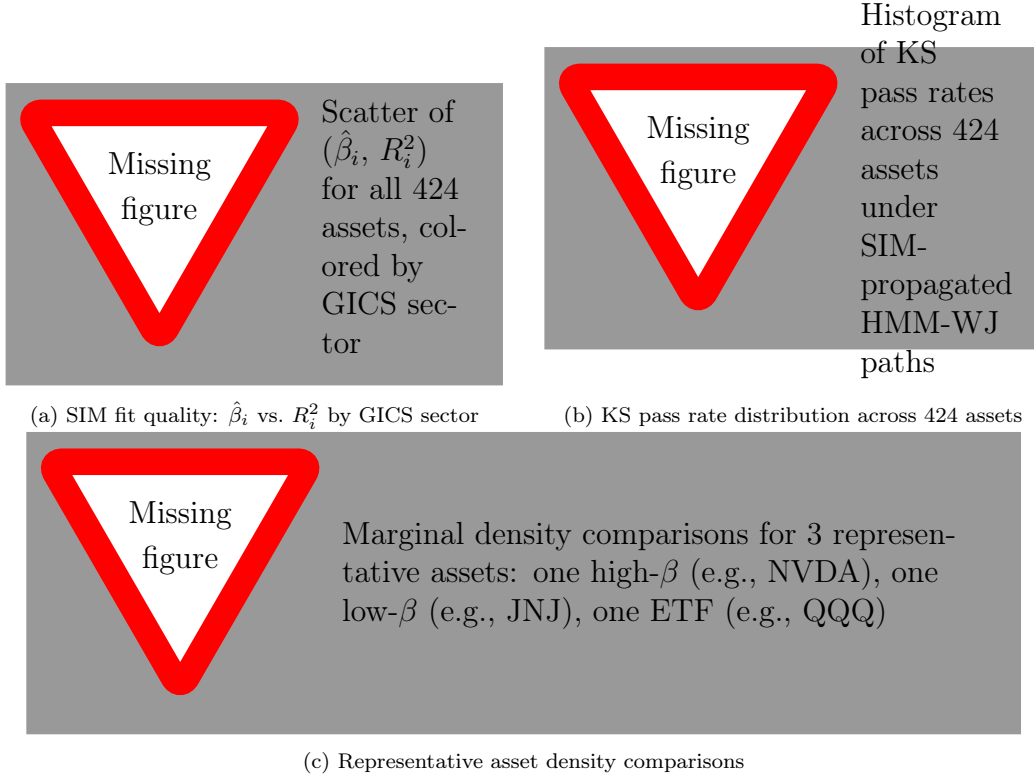


Figure 7: Multi-asset extension via the Single-Index Model (SIM) across 424 S&P 500 constituents. Panel (a) summarises the SIM regression fit quality by GICS sector. Panel (b) shows that KS distributional consistency is maintained across the full asset universe when paths are generated via the factor structure $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{\text{SPY},t} + \hat{\eta}_{i,t}$. Panel (c) illustrates representative marginal density comparisons for three assets spanning a wide range of systematic risk exposure.

Table 1: Descriptive statistics and stylized-facts tests for SPY daily excess growth rates (in-sample period 2014–2024; $T = 2,766$ trading days). JB = Jarque-Bera normality test. LB = Ljung-Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	SPY Observed	Gaussian	Laplace
Mean (annualized, %)	—	—	—
Std Dev (annualized, %)	—	—	—
Skewness	—	0	0
Excess Kurtosis	—	0	3
JB normality test	reject*	—	—
LB test on G_t (lag 20)	fail to reject	—	—
LB test on $ G_t $ (lag 20)	reject*	—	—

Table 2: In-sample and out-of-sample model comparison for SPY ($N = 100$, 1,000 simulated paths, significance level $\alpha = 0.05$). GARCH: GARCH(1,1) estimated by quasi-maximum likelihood. HMM-NJ: standard Hidden Markov Model without jumps. HMM-WJ: hybrid model with Poisson jump-duration extension. ACF-MAE: mean absolute error of the ACF of $|G_t|$ over lags 1–252. Kurtosis: mean excess kurtosis across simulated paths. Dashes indicate values pending computation.

Metric	GARCH(1,1)	HMM-NJ ($N = 100$)	HMM-WJ ($N = 100$)
<i>In-sample: 2,766 trading days (2014–2024)</i>			
KS pass rate (%)	6.0	100.0	96.4
AD pass rate (%)	1.4	97.0	89.8
Excess kurtosis (observed)		7.715	
Excess kurtosis (simulated)	8.1	5.7	5.5
ACF-MAE (lags 1–252)	0.030	0.059	0.052
<i>Out-of-sample: 249 trading days (2025)</i>			
KS pass rate (%)	78.1	96.6	94.6
AD pass rate (%)	73.1	96.7	95.2
Excess kurtosis (observed)		6.867	
Excess kurtosis (simulated)	1.6	5.1	4.9
ACF-MAE (lags 1–252)	0.026	0.041	0.039
Parameters estimated	3	2	4

Table 3: Sensitivity of model performance to state resolution N . KS and AD pass rates computed over 1,000 simulated in-sample paths at $\alpha = 0.05$. ACF-MAE = mean absolute error of the ACF of $|G_t|$ over lags 1–252 (pending).

Model	N	KS pass (%)	AD pass (%)	ACF-MAE
HMM-NJ	90	99.4	98.4	—
HMM-NJ	60	99.4	98.3	—
HMM-NJ	30	99.2	98.2	—
HMM-WJ	90	99.5	98.1	—
HMM-WJ	60	99.3	97.6	—
HMM-WJ	30	98.7	97.7	—

Table 4: Multi-asset extension results via the Single-Index Model (SIM) across 424 S&P 500 constituents. Paths generated as $\hat{G}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i G_{\text{SPY},t} + \hat{\eta}_{i,t}$ where $G_{\text{SPY},t}$ is drawn from HMM-WJ ($N = 100$) simulated paths. All statistics pending completion of the 424-asset pipeline.

Metric	Mean	Median	5th pct	95th pct
KS pass rate (%)	—	—	—	—
AD pass rate (%)	—	—	—	—
SIM R^2	—	—	—	—
Excess kurtosis match	—	—	—	—

Algorithms

Algorithm 1 Empirical hidden Markov model construction via Laplace partitioning

Require: Price history $P_{1:T}$, Risk-free rate r_f , Number of states N

Ensure: Transition matrix $\mathbf{T}$, Quantile boundaries $\mathcal{Q}$

- 1: Compute excess growth rates $R_t = \frac{1}{\Delta t} \ln(P_t/P_{t-1}) - r_f$ for all t .
 - 2: Fit Laplace parameters (μ_L, b_L) to the series $\mathbf{R}$ using maximum-likelihood estimation.
 - 3: Define interior boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N-1\}$, where F_L^{-1} is the inverse CDF of the Laplace distribution.
 - 4: Set finite outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$.
 - 5: Assign each excess growth rate R_t to a discrete state $s_t \in \{1, \dots, N\}$ based on the boundaries in $\mathcal{Q} = \{Q_k\}$.
 - 6: Initialize count matrix $\mathbf{C} \in \mathbb{R}^{N \times N}$ and transition matrix $\mathbf{T} \in \mathbb{R}^{N \times N}$ with zeros.
 - 7: **for** $t = 1$ to $T - 1$ **do**
 - 8: Identify transition $i = s_t$ to $j = s_{t+1}$.
 - 9: $\mathbf{C}_{i,j} \leftarrow \mathbf{C}_{i,j} + 1$.
 - 10: **end for**
 - 11: **for** $i = 1$ to N **do**
 - 12: $\mathbf{T}_{i,:} \leftarrow \mathbf{C}_{i,:} / \sum_{j=1}^N \mathbf{C}_{i,j}$ {Row-wise normalization}
 - 13: **end for**
 - 14: **return** $\mathbf{T}, \mathcal{Q}$
-

Algorithm 2 Hybrid jump-diffusion simulation with persistent volatility

Require: Model parameters $\mathbf{T}, \bar{\pi}, \epsilon, \lambda, N_{tail}, p_{neg}$, Total Steps M

Ensure: Simulated state sequence $S_{1:M}$

```
1: Define tail-states  $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$  and  $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$ .
2: Sample initial state  $S_1 \sim \text{Categorical}(\bar{\pi})$ .
3: Set  $counter \leftarrow 2$ .
4: while  $counter \leq M$  do
5:   Sample  $u \sim \text{Uniform}(0, 1)$ .
6:   if  $u < \epsilon$  then
7:     Sample jump-duration  $K \sim \text{Poisson}(\lambda)$ .
8:     for  $j = 1$  to  $K$  while  $counter \leq M$  do
9:       Sample  $w \sim \text{Uniform}(0, 1)$ .
10:      if  $w < p_{neg}$  then
11:         $S_{counter} \sim \text{Uniform}(\mathcal{S}_{bottom})$  {Bias toward negative-tail events}
12:      else
13:         $S_{counter} \sim \text{Uniform}(\mathcal{S}_{top})$ 
14:      end if
15:       $counter \leftarrow counter + 1$ .
16:    end for
17:  else
18:    Sample  $S_{counter} \sim \text{Categorical}(\mathbf{T}_{S_{counter-1}, :})$ .
19:     $counter \leftarrow counter + 1$ .
20:  end if
21: end while
22: return  $S_{1:M}$ 
```

Algorithm 3 State decoding and continuous excess growth rate reconstruction

Require: Simulated state sequence $S_{1:M}$, Quantile boundaries $\mathcal{Q}$

Ensure: Reconstructed continuous excess growth rates $\hat{R}_{1:M}$

- 1: Define $z_{0.999} = 3.09$.
 - 2: **for** $k = 1$ to N **do**
 - 3: Set $\mu_k = (Q_{k-1} + Q_k)/2$.
 - 4: Set $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$.
 - 5: **end for**
 - 6: Initialize the reconstructed excess growth rate vector $\hat{\mathbf{R}}$ of length M .
 - 7: **for** $t = 1$ to M **do**
 - 8: Retrieve the simulated state $k = S_t$.
 - 9: Sample the continuous excess growth rate $\hat{R}_t$ from the normal distribution $\mathcal{N}(\mu_k, \sigma_k)$.
 - 10: **end for**
 - 11: **return** $\hat{R}_{1:M}$
-

Supplemental Information

Algorithm 4 Multi-objective grid search for jump hyperparameters

Require: Observed growth rates $\mathbf{R}_{obs}$, ACF max lag $L = 252$, Number of paths $P = 1000$, Time steps $M = 2766$

Require: Grid space for $\epsilon \in [10^{-5}, 10^{-4}]$ and $\lambda \in [1.0, 100.0]$

Require: Kurtosis penalty weight $w_K = 0.20$

Ensure: Optimal jump hyperparameters (ϵ^*, λ^*)

- 1: Calculate observed absolute ACF: $A_{obs}(\tau) = \text{ACF}(|\mathbf{R}_{obs}|, \tau)$ for $\tau \in \{1, \dots, L\}$.
 - 2: Calculate observed global kurtosis: $K_{obs} = \text{Kurtosis}(\mathbf{R}_{obs})$.
 - 3: Initialize minimum error $E_{min} \leftarrow \infty$.
 - 4: **for** each ϵ in grid space **do**
 - 5: **for** each λ in grid space **do**
 - 6: Initialize simulated ACF accumulator $\mathbf{A}_{sum} \leftarrow \mathbf{0}$ and kurtosis accumulator $K_{sum} \leftarrow 0$.
 - 7: **for** $p = 1$ to P **do**
 - 8: Simulate path using Algorithm 2 to obtain state sequence $S_{1:M}$.
 - 9: Decode sequence using Algorithm 3 to obtain continuous rates $\hat{\mathbf{R}}^{(p)}$.
 - 10: $\mathbf{A}_{sum}(\tau) \leftarrow \mathbf{A}_{sum}(\tau) + \text{ACF}(|\hat{\mathbf{R}}^{(p)}|, \tau)$ for all τ .
 - 11: $K_{sum} \leftarrow K_{sum} + \text{Kurtosis}(\hat{\mathbf{R}}^{(p)})$.
 - 12: **end for**
 - 13: Compute ensemble averages: $\bar{A}_{sim}(\tau) = \mathbf{A}_{sum}(\tau)/P$ and $\bar{K}_{sim} = K_{sum}/P$.
 - 14: Compute objective: $J = \sum_{\tau=1}^L (A_{obs}(\tau) - \bar{A}_{sim}(\tau))^2 + w_K (K_{obs} - \bar{K}_{sim})^2$.
 - 15: **if** $J < E_{min}$ **then**
 - 16: $E_{min} \leftarrow J$
 - 17: $(\epsilon^*, \lambda^*) \leftarrow (\epsilon, \lambda)$
 - 18: **end if**
 - 19: **end for**
 - 20: **end for**
 - 21: **return** ϵ^*, λ^*
-